

Question # 1:-

Marks = 30 [2+25+3]

a) ^② Response time = End time - Start time

System A
Response time

①

$$118 - 102 = 16$$

$$130 - 110 = 20$$

$$140 - 115 = 25$$

$$150 - 120 = 30$$

$$145 - 125 = 20$$

$$155 - 130 = 25$$

$$160 - 135 = 25$$

$$165 - 140 = 25$$

$$175 - 145 = 30$$

$$190 - 150 = 40$$

System B
Response time

①

$$210 - 200 = 10$$

$$220 - 205 = 15$$

$$230 - 210 = 20$$

$$240 - 215 = 25$$

$$250 - 220 = 30$$

$$255 - 225 = 30$$

$$260 - 230 = 30$$

$$270 - 235 = 35$$

$$275 - 240 = 35$$

$$310 - 245 = 65$$

b) ^{②⑤} Five - point summary for System A.

Sorted data : 16, 20, 20, 25, 25, 25, 25, 30, 30, 40

Minimum = 16 ①

For Q_1 : $i = \left(\frac{25}{100}\right) \times 10 = 2.5^{th}$ index so round up
hence $Q_1 = 20$ ①

For Q_2 : $i = \left(\frac{50}{100}\right) \times 10 = 5^{th}$ index so Median = $\frac{5^{th} + 5^{th}}{2}$

$$Q_2 = \frac{25 + 25}{2} = 25 \quad \text{hence } \boxed{Q_2 = 25} \quad ①$$

For Q_3 : $i = \left(\frac{75}{100}\right) \times 10 = 7.5^{th}$ index so round up

hence $Q_3 = 30$ ①

Maximum = 40 ①

Five point Summary for System B

pg ②

Sorted data: 10, 15, 20, 25, 30, 30, 30, 35, 35, 65

$$\boxed{\text{Minimum} = 10} \quad ①$$

For Q_1 , 2.5^{th} index hence $\boxed{Q_1 = 20} \quad ①$

For Q_2 , $(\frac{50}{100}) \times 10 = 5^{\text{th}}$ index hence $Q_2 = \frac{5^{\text{th}} + 6^{\text{th}}}{2}$

$$Q_2 = \frac{30 + 30}{2}, \quad \boxed{Q_2 = 30} \quad ①$$

For $Q_3 = (\frac{75}{100}) \times 10 = 7.5^{\text{th}}$ index hence $Q_3 = 8^{\text{th}}$ value

$$\boxed{Q_3 = 35} \quad ①$$

$$\boxed{\text{Maximum} = 65} \quad ①$$

Outlier Detection:-

System A

$$\begin{aligned} \text{Lower bound} &= Q_1 - 1.5(Q_3 - Q_1) \\ &= 20 - 1.5(30 - 20) \\ &= 20 - 15 = 5 \quad ① \end{aligned}$$

$$\begin{aligned} \text{upper bound} &= Q_3 + 1.5(Q_3 - Q_1) \\ &= 30 + 1.5(30 - 20) \\ &= 45 \quad ① \end{aligned}$$

No outlier in System A. ①
No significant bottlenecks.

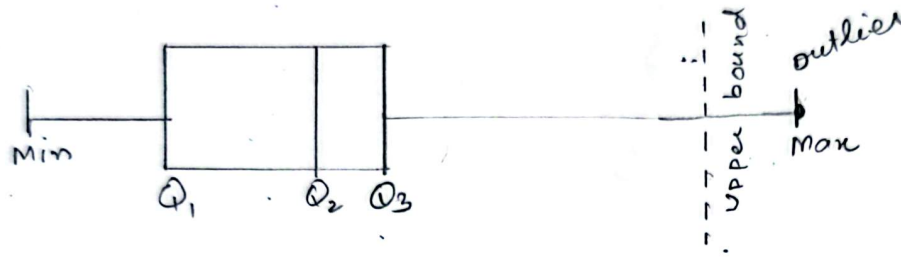
System B

$$\begin{aligned} \text{Lower bound} &= 20 - 1.5(35 - 20) \\ &= 20 - 1.5(15) \\ &= -2.5 \quad ① \end{aligned}$$

$$\begin{aligned} \text{upper bound} &= 35 + 1.5(35 - 20) \\ &= 35 + 1.5(15) \\ &= 57.5 \quad ① \end{aligned}$$

65 is an outlier, which shows occasional performance bottlenecks. ①

Lower bound



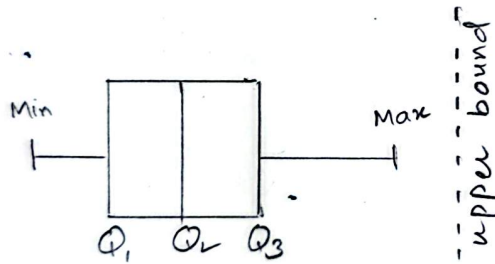
③

System B

①

For outlier

Lower bound



③

System A

①

for scale

Skewness:-

System A is approximately symmetric. ①

System B is Negatively Skewed. (Left Skewed)

[Notes: If any student use quantitative measure then results are negatively skewed for system B, But if comment directly through box-plot whiskers then it is positively skewed.]

③

C) IQR (System A)

$$IQR = Q_3 - Q_1 = 30 - 20 = 10 \quad ①$$

IQR (System B)

$$IQR = Q_3 - Q_1 = 35 - 20 = 15 \quad ①$$

Hence it shows that system A demonstrates more consistent response time performance than system B. ①

Question # 1 (Second Method): Marks = 30 [2+25+7] Pg 5

a) $S_A: 16 \quad 20 \quad 25 \quad 30 \quad 20 \quad 25 \quad 25 \quad 25 \quad 30 \quad 40$
 $S_B: 10 \quad 15 \quad 20 \quad 25 \quad 30 \quad 30 \quad 30 \quad 35 \quad 35 \quad 65$ (2)

b) Five number Summary

$S_A: 16 \quad 20 \quad 20 \quad 25 \quad 25 \quad 25 \quad 25 \quad 30 \quad 30 \quad 40$
 $S_B: 10 \quad 15 \quad 20 \quad 25 \quad 30 \quad 30 \quad 30 \quad 30 \quad 35 \quad 35 \quad 65$

Min Σ S_A 16

$$Q_1: \frac{(10+1)}{4}^{th} = 2.75^{th}$$

$$Q_1 = 20 + 0.75(20 - 20) = 20$$

$$Q_2: 2 \left(\frac{10+1}{4} \right)^{th} = 5.5^{th}$$

$$Q_2 = 25 + 0.50(25 - 25) = 25$$

$$Q_3: 3 \left(\frac{10+1}{4} \right)^{th} = 8.25^{th}$$

$$= 30 + 0.25(30 - 30) = 30$$

$$\text{Max} = 40$$

$$\text{Lower limit} = Q_1 - 1.5 IQR$$

$$= 20 - 1.5(30 - 20) = 5$$

$$\text{Upper limit} = Q_3 + 1.5 IQR$$

$$= 30 + 1.5(30 - 20) = 45$$

Min Σ 10

$$Q_1 = 15 + 0.75(20 - 15)$$

$$= 18.75$$

$$Q_2 = 30 + 0.5(30 - 30)$$

$$= 30$$

$$Q_3 = 35 + 0.25(35 - 35)$$

$$= 35$$

$$\text{Max} = 65$$

Lower limit

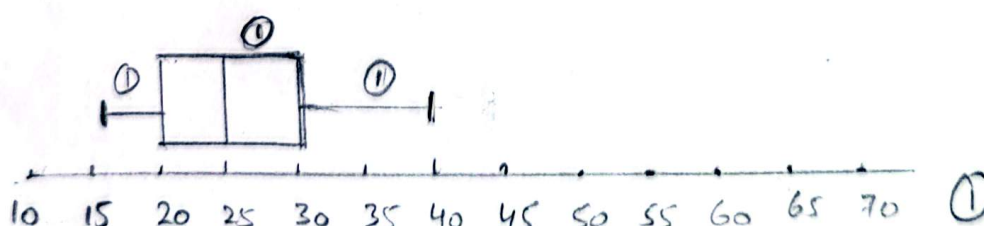
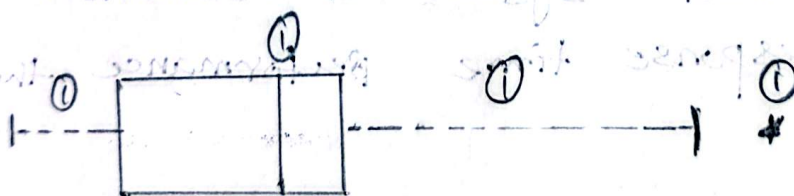
$$= 18.75 - 1.5(35 - 18.75)$$

$$= -5.625$$

Upper limit =

$$35 + 1.5(35 - 18.75)$$

$$= 59.375$$



Outlier detection:-

Pg-6

No outlier in System A. System B suggests the presence of an occasional performance bottleneck due to outlier 65

Shape: Both the distributions are positively skewed and indicates most requests are processed efficiently but System B is strongly positively skewed because of 65ms outlier indicates more serious occasional slowdowns.

③ C) $IQR = Q_3 - Q_1$
System A

$$IQR = (30 - 20) \\ = 10$$

System B

$$IQR = 35 - 18.75 \\ = 16.25$$

So System A is more consistent than System B.

Question # 2

Marks = 15 [5+5+2+2+1]

⑤ a)

$y \backslash x$	2	3	4	$h(y)$
6	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{3}$
7	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{3}$
8	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{3}$
$g(x)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

⑤
b)

probability distribution of T

T	f(t)
8	1/9
9	2/9
10	3/9
11	2/9
12	1/9
	1

⑤

②
c)

$$E(T) = \sum t f(t)$$

$$= (8 \times \frac{1}{9}) + (9 \times \frac{2}{9}) + (10 \times \frac{3}{9}) + (11 \times \frac{2}{9}) + (12 \times \frac{1}{9}) \quad \text{①}$$

$$= \frac{8}{9} + \frac{18}{9} + \frac{30}{9} + \frac{22}{9} + \frac{12}{9} = \frac{90}{9}$$

$$E(T) = 10$$

①

②

$$d) \sum t^2 f(t) = \frac{64}{9} + \frac{162}{9} + \frac{300}{9} + \frac{242}{9} + \frac{144}{9} = \frac{912}{9} \quad \text{①}$$

$$S.D(t) = \sqrt{\sum t^2 f(t) - [\sum t f(t)]^2}$$

$$= \sqrt{\frac{912}{9} - (10)^2} = \sqrt{101.333 - 100}$$

$$S.D(t) = \sqrt{1.333} = 1.154$$

①

①
e)

$$P(T \leq 10) = \frac{1}{9} + \frac{2}{9} + \frac{3}{9}$$

$$= \frac{6}{9} = \frac{2}{3}$$

①

Question # 3a

Marks = 10 [4+6] Pg-8

i) $\mu = 12, \sigma = 3$

①

$$P(X > 10) = 1 - \Phi\left(\frac{10-12}{3}\right) = 1 - \Phi(-0.67) \quad (1)$$

$$= 1 - 0.2514$$

$$= 0.7486 \quad (1)$$

ii) $P(X > a) = 0.05$ ①

$$0.05 = 1 - \Phi\left(\frac{a-12}{3}\right) = \Phi\left(\frac{a-12}{3}\right) = 0.95 \quad (1)$$

$$\frac{a-12}{3} = \Phi^{-1}(0.95)$$

②

$$\frac{a-12}{3} = 1.645$$

$$a = 12 + (3 \times 1.645)$$

②

$$\boxed{a = 16.935}$$

Question # 3b

Marks = 10 [5+5]

⑤ i) The number of packet arrival per minute follows the poisson distribution. As mean and variance are equal in poisson distribution so

$$\lambda t = 1.8 \text{ packet per minute}$$

$$f(x) = \frac{e^{-1.8} (1.8)^x}{x!}, \quad x = 0, 1, 2, \dots \quad (1)$$

$$P(\text{congestion}) = P(X > 2) \Rightarrow 1 - P(X \leq 2)$$

$$= 1 - \{P(X=0) + P(X=1) + P(X=2)\}$$

$$= 1 - \left\{ \frac{e^{-1.8} (1.8)^0}{0!} + \frac{e^{-1.8} (1.8)^1}{1!} + \frac{e^{-1.8} (1.8)^2}{2!} \right\} \quad (1)$$

$$= 1 - \{0.1653 + 0.2975 + 0.2678\} \quad (2)$$

$$= 1 - 0.7306 = 0.2694 \quad (1)$$

(ii) Average number of packet arrival per 15 second interval is Pg - 9

$$\lambda t = 1.8 \times \frac{15}{60} = 0.45 \quad (1)$$

$$P(X > 3) = 1 - P(X \leq 3) = 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)]$$

$$= 1 - \left[e^{-0.45} \frac{(0.45)^0}{0!} + e^{-0.45} \frac{(0.45)^1}{1!} + e^{-0.45} \frac{(0.45)^2}{2!} + e^{-0.45} \frac{(0.45)^3}{3!} \right] \quad (2)$$

$$= 1 - [0.6376 + 0.2869 + 0.0645 + 0.0097] \quad (2)$$

$$= 1 - 0.9988$$

$$= 0.0012 \quad (1)$$

Question # 4:-

Marks = 10

$n = 40$, for known σ the 95% C-I for μ ($\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$) is 0.30 - 0.50 (1)

$$\alpha = 1 - C.C = 1 - 0.95 = 0.05$$

$$Z_{\frac{\alpha}{2}} = Z_{0.05/2} = 1.96 \quad (1)$$

$$\bar{x} - 1.96 \frac{\sigma}{\sqrt{40}} = 0.30 \quad \text{equation 1} \quad (1)$$

$$\bar{x} + 1.96 \frac{\sigma}{\sqrt{40}} = 0.50 \quad \text{eq 2}$$

By Adding equation (1) and equation (2) we have $2\bar{x} = 0.80 \Rightarrow \bar{x} = 0.40$, put $\bar{x} = 0.40$ in eq (2)

$$0.40 + 1.96 \frac{\sigma}{\sqrt{40}} = 0.50 \Rightarrow \sigma = \frac{0.10}{0.3099} \Rightarrow 0.323$$

Now $n = 40$, $\bar{x} = 0.40$, $\sigma = 0.323$ and $C.C = 1 - 0.01 = 0.99$ (1)

$$99\% \text{ C-I for mean } \Rightarrow \bar{x} \pm 2.575 \frac{\sigma}{\sqrt{n}} \Rightarrow 0.40 \pm 2.575 \left(\frac{0.323}{\sqrt{40}} \right)$$

$$0.40 - 0.1315 < \mu < 0.40 + 0.1315$$

$$0.2685 < \mu < 0.5315 \quad (2)$$

Question # 5:-

$$\hat{y} = 9.82 + 2.4955x$$

Pg 10

Marks = 9+16=25

X	Y	X Y	X ²	Y ²
1	12	12	1	144
2	15	30	4	225
3	18	54	9	324
3	17	51	9	289
4	20	80	16	400
5	23	115	25	529
5	21	105	25	441
6	25	150	36	625
7	27	189	49	729
8	30	240	64	900
	208	1026	238	4606

Q

a)

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum (Y - \hat{Y})^2}{\sum (Y - \bar{Y})^2}$$

$$= 1 - \frac{(\sum Y^2 - b_0 \sum Y - b_1 \sum XY)}{\sum Y^2 - (\sum Y)^2/n}$$

$$= 1 - \frac{4606 - (9.82)(208) - (2.4955)(1026)}{4606 - (208)^2/10}$$

$$= 1 - \frac{4606 - 2042.56 - 2560.383}{4606 - 4326.4}$$

$$= 1 - \frac{3.057}{279.6} = 1 - 0.0109$$

$$= 0.989$$

$$SST = 3 \text{ Marks}$$

$$SSE = 3 \text{ Marks}$$

2

Interpretation:-

98.9% of the variation in ①

(Y) dependent variable (Inference time) is explained by its linear relationship with (X) independent variable (complexity level). 1.1% variation is due to chance or other factors

b) $SST = \sum Y^2 - \frac{(\sum Y)^2}{n} = 4606 - \frac{(208)^2}{10} = 279.6$ Pg-11

$$SSE = \sum Y^2 - b_0 \sum Y - b_1 \sum XY$$

$$= 4606 - (9.82 \times 208) - (2.4955 \times 1026)$$

$$= 3.057$$

$$SSR = SST - SSE$$

$$= 279.6 - 3.057$$

$$= 276.543$$

ANOVA TABLE					
S.O.V	Sum of Squares	D. f	Mean Squares	F-Statistic	P-value
Regression	SSR = 276.543	1	MSR = 276.543	$F = \frac{MSR}{MSE} = \frac{276.543}{0.3822} = 723.55$	0.0001
Error	SSE = 3.057	n-2 = 8	$MSE = \frac{3.057}{8} = 0.3822$		
Total	.	n-1 = 9			

Testing Procedure:-

- 1) H_0 : Regression is insignificant
 H_1 : Regression is significant.

2) $\alpha = 0.05$

3) $F = \frac{MSR}{MSE} = \frac{276.543}{0.3822} = 723.55$

4) Since $p\text{-value} < \alpha$
 $0.0001 < 0.05$

5) Decision: Reject H_0 .
 Conclusion: overall regression model is significant.

Note: If any student uses formulas by definition:

$$SSE = \sum (Y - \hat{Y})^2 = 3.10$$

$$SSR = \sum (\hat{Y} - \bar{Y})^2 = 276.50$$

$$SST = \sum (Y - \bar{Y})^2 = 279.60$$

These small difference are due to estimated values.